

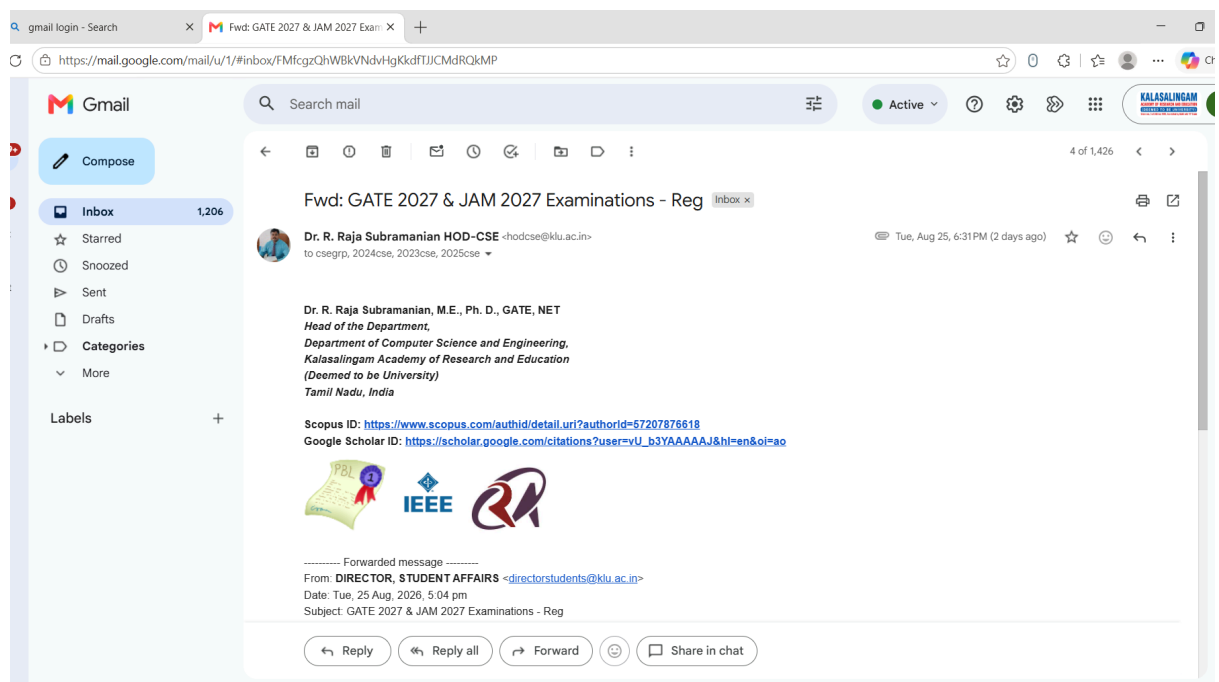
Ex. No 4: Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)

Description:

Here's a step-by-step guide on how to execute and analyze an email header:

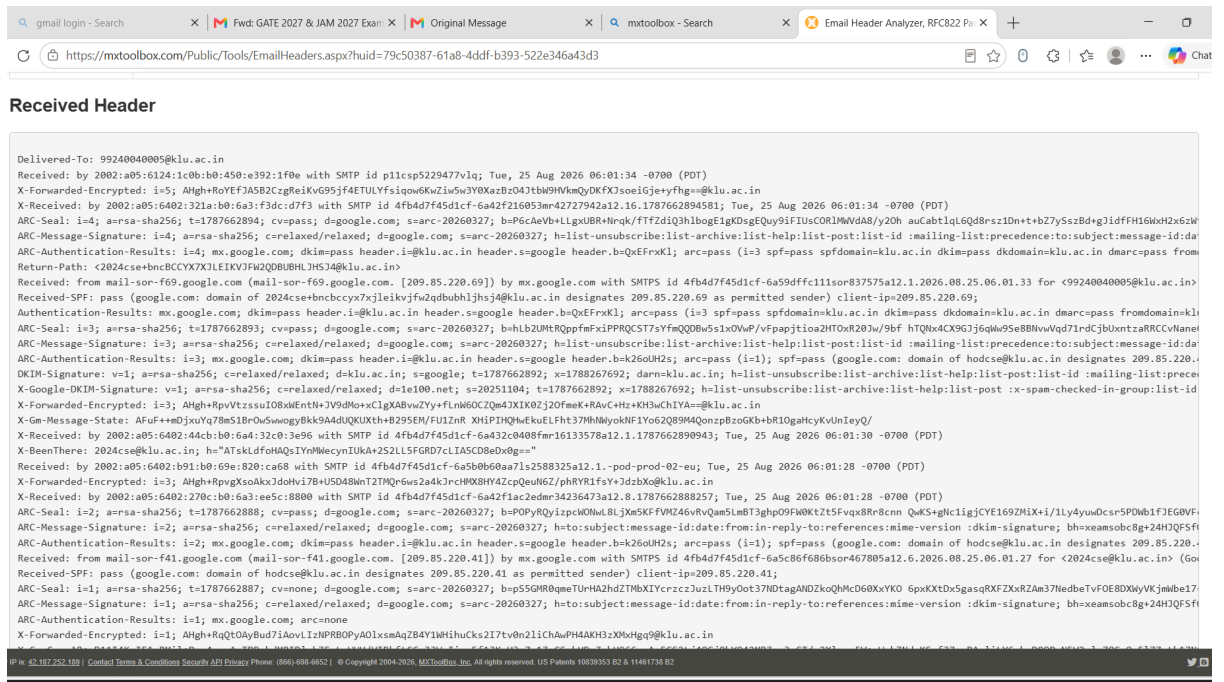
1. Access the Email Header

- **Gmail:** Open the email, click the three dots (More) in the upper right corner, and select "Show original."
- **Outlook:** Open the email, click on "File," then "Properties," and find the "Internet headers" box.
- **Yahoo:** Open the email, click on the three dots (More), and select "View raw message."



2. Copy the Email Header

- Once you access the header, copy all the text that appears. This text contains various metadata fields about the email's journey from the sender to your inbox.



3. Identify Key Header Fields

- **From:** The sender's email address.
- **To:** The recipient's email address.
- **Date:** The date and time the email was sent.
- **Subject:** The email's subject line.
- **Return-Path:** The return address if the email bounces.
- **Received:** A list of servers the email passed through on its way to the recipient. This is usually the most critical part of header analysis.
- **Message-ID:** A unique identifier for the email.
- **SPF (Sender Policy Framework) / DKIM (DomainKeys Identified Mail):** Indicators of email authentication.

Original Message

Message ID	<CAMq6mVb2KZH-MSBe-cAKSiLSC1EU41MSv0o6e5-j5=cObjpLXg@mail.gmail.com>
Created at:	Tue, Aug 25, 2026 at 6:31 PM (Delivered after 22 seconds)
From:	"Dr. R. Raja Subramanian HOD-CSE" <hodcse@klu.ac.in>
To:	csegrp@klu.ac.in, 2024cse@klu.ac.in, 2023cse@klu.ac.in, 2025cse@klu.ac.in
Subject:	Fwd: GATE 2027 & JAM 2027 Examinations - Reg
SPF:	PASS with IP 209.85.220.69 Learn more
DKIM:	'PASS' with domain klu.ac.in Learn more
DMARC:	'PASS' Learn more

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```

Delivered-To: 99240040005@klu.ac.in
Received: by 2002:a05:6124:1c0b:b0:450:e392:1f0e with SMTP id p11csp5229477v1q; Tue, 25 Aug 2026 06:01:34 -0700 (PDT)
X-Forwarded-Encrypted: i=5; AHgh+RoYEFJA5B2CzgreiKvG95jT4ETULYfsIqow6KwZ1w5w3Y0XaZ04JtbW9HVKmQyDKfXJsoe1Gj+eyfhg==@klu.ac.in
X-Received: by 2002:a05:6402:321a:b0:6a3:f3dc:d7f3 with SMTP id 4fb4d7f45d1cf-6a42f216053mr42727942a12.16.1787662894581; Tue, 25 Aug 2026 06:01:34 -0700 (PDT)
ARC-Seal: i=4; a=rsa-sha256; t=1787662894; cv=pass; d=google.com; s=arc-20260327; b=P6cAeVb+LLgxUBR+HrQk/FTfZdiQ3h1bogE1gKDsGEQuY9iFIUSCORJMWdA8/y20h auCabtlQL6Qd8rsz1Dn+t+bZ7ySszBd+g3ldffH16WxH2x6ZwFUK+HCHKJeq7jR3tRDK k5Dn1M5RVvvvT010A4Rm<56FH161TMBH+7YVkvQvF0MNDHF:a0TXuVfui m1cR8Ph2mkamH iHS4XXVamFKK1/c10xT6vFnuGr1ANDSw71nVnRav0/iX7D6J273K0FvnrTDN3iPSBw6

```

4. Analyze the 'Received' Fields

- The Received fields show the path the email took from the sender to the recipient. These fields are listed in reverse order (from the last server to the first).
- Each Received line will show:
 - The sending server's hostname/IP address.
 - The receiving server's hostname/IP address.
 - The date and time the email passed through.

5. Check for IP Addresses and Hostnames

- Use tools like **WHOIS** or online IP lookup services to identify the geographical location and ownership of the IP addresses found in the Received lines.
- Check if any IP addresses are suspicious or if the hostname does not match the expected sending server.

Whois IP 209.85.220.69

Updated 1 day ago

```
#
# ARIN WHOIS data and services are subject to the Terms of Use
# available at: https://www.arin.net/resources/registry/whois/tou/
#
# If you see inaccuracies in the results, please report at
# https://www.arin.net/resources/registry/whois/inaccuracy_reporting/
#
# Copyright 1997-2026, American Registry for Internet Numbers, Ltd.
#

NetRange: 209.85.128.0 - 209.85.255.255
CIDR: 209.85.128.0/17
NetName: GOOGLE
NetHandle: NET-209-85-128-0-1
Parent: NET209 (NET-209-0-0-0-0)
NetType: Direct Allocation
OriginAS:
Organization: Google LLC (GOGL)
RegDate: 2006-01-13
Updated: 2012-02-24
Ref: https://rdap.arin.net/registry/ip/209.85.128.0

OrgName: Google LLC
OrgId: GOGL
Address: 1600 Amphitheatre Parkway
City: Mountain View
State: CA
PostalCode: 94043
Country: US
```

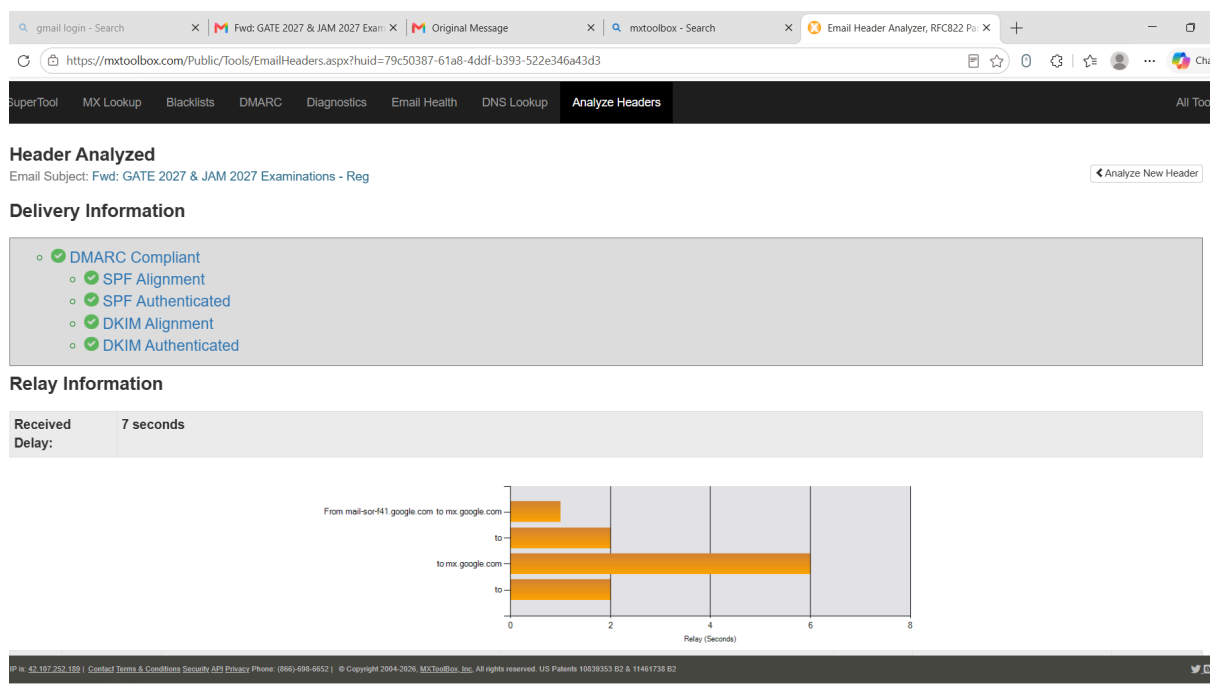
.space Sale
\$1.18
BUY NOW
*while stocks last

On Sale!
.info
\$.INFO @ \$5.28 \$32.88

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Build A Beautiful Website In Minutes!

6. Examine the SPF, DKIM, and DMARC Results

- **SPF:** Checks if the sender's IP is authorized to send emails for the domain.
 - **Pass:** Indicates the IP is authorized.
 - **Fail:** The IP is not authorized and could be a sign of spoofing.
- **DKIM:** Verifies that the email content has not been altered in transit.
 - **Pass:** The email has not been tampered with.
 - **Fail:** The email content may have been altered.
- **DMARC:** A policy that uses SPF and DKIM to authenticate the email.
 - **Pass:** Indicates alignment with SPF/DKIM policies.
 - **Fail:** Indicates possible spoofing or tampering.



7. Analyze the 'Message-ID'

- The Message-ID is unique to each email. It usually includes the domain of the sending server.
- Check if the domain in the Message-ID matches the sender's domain. If not, it could be a sign of spoofing.

8. Look for Anomalies

- **Mismatch in Email Domains:** Check if the From domain matches the domains in Return-Path, Message-ID, and other authentication results.
- **Strange IP Addresses:** Be wary of IP addresses that do not match the known sending server or are from unexpected locations.
- **Suspicious Time Stamps:** Check for significant time discrepancies in the Received lines, which could indicate delays or rerouting.

9. Use Online Tools

- Use online email header analysis tools like **MXToolbox** or **Google's G Suite Toolbox** to automatically parse and analyze the email header.

gmail login - Search

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Original Message

mxtoolbox - Search

Email Header Analyzer, RFC822 P...

https://mxtoolbox.com/Public/Tools/EmailHeaders.aspx?huid=79c50387-61a8-4ddf-b393-522e346a43d3

dkim:klu.ac.in:google

Show

Dkim Public Record:

v=Dkim1; k=rsa; p=MIIBIjANBgkqhkiG9w0BAQEFAAACQ8AMIIBCBgKCAQEAqsFm7ayERMKQJAd63IK15oq14du/BKVZYlyueFTX5nhQ12E03LrPwQeYyNs7oNbpane02c5D9r7x1XC5sEd3X16p5am44TCmP/5gLVaGDDa11P84qh45HhBHEsm1q+

Dkim Signature:

v=1; a=rsa-sha256; c=relaxed/relaxed; d=klu.ac.in; s=google; t=1787662892; x=1788267692; darn=klu.ac.in; h=list-unsubscribe:list-archive:list-help:list-post:list-id :mailing-list:precedence:x

Headers Found

Header Name	Header Value
Delivered-To	99240040005@klu.ac.in
X-Forwarded-Encrypted	i=5, AHgh+RoYEUA5B2CzgReiKvG95j4ETULYfsiqow6KwZw5w3Y0XazBzO4JtbW9HvkmQyDKDJsociGje+yfhg==@klu.ac.in
X-Received	by 2002.a05.6402.321a.b0.6a3.f3dc.d7f3 with SMTP id 4fb4d7f45d1cf6a42f216053mr42727942a12.16.1787662894581; Tue, 25 Aug 2026 06:01:34 -0700 (PDT)
ARC-Seal	i=4, a=rsa-sha256, c=relaxed/relaxed, d=google.com, s=arc-20260327, b=P6cAeVb+LLgxUBR+NrgkITIZdIQ3hbogE1gKDsgEQuy9fIUscCORIMWVdA8y2Oh auCabtlqL6Od8srz1Dn+!+bZ7ySszBd+qJidffH16WxH2x6zWUUK+WcHKleq7JR3IRDk kSPqjMSRVxvTQID448msS6EHBJNNHh7YYKy9yfgNQDHFa0UXyNVWMLCg8PbzmaMlj iHS4XXVaWFKXj/sIOxTByfGvc14NDSW71nVnQ0yQLX7P6J42JK0tyDrTPNiP5Bw6 ma4OUmI9+YJs5ph72GYufK7dDC/ChOlQs4OyMgb642Eh9z3g4emtJL7UapchcnS iBMg==
ARC-Message-Signature	i=4, a=rsa-sha256, c=relaxed/relaxed, d=google.com, s=arc-20260327, h=list-unsubscribe:list-archive:list-help:list-post:list-id :mailing-list:precedence:subject:message-id:date:from:in-reply-to:references:mime-version:dkim-signature; bh=xamsobcbg+24HJQFSIO9vaVHnkH49/GRizniCuWAaM=-; fh=0HOTGELYKJw07MZYfWwWtAES2+y6R7PQdHCcnemc-; b=blWqHthOMaNNVwJy0XPPA1DxJZTVwMVyMIRGEaLyQIEN0IHdZeZfoOsXyWdyukE 7gT6B/TmRWmo9S9Eaz7Rgzh8XUhrBj6q7I3Hl+J3Jz4agpWS6J29vp2VAA08FNknsMt Xn6y7vRfRy68q+IM7GauTF6wbB5wvleuF5BvECOLBzOOQ+6Bm4s/xan5FQmQSVzHD5s5aC oyufZ85ERasYdPvBRY1vX9RiXzZd6DdThU/AjU+brHLcs.JJoQSPHRg9f70yYYDle RdoEQOKOTVTIRz521D0GI4XOWQ5mEnySFSdQbJJsudGQNMAYVpM+f8Rcb9KZgva DMYW=-; dara=google.com
ARC-Authentication-Results	i=4, mx.google.com, dkim=pass header i=@klu.ac.in header s=google header b=QxEfXkI; arc=pass (i=3 spf=pass spfdomain=klu.ac.in dkim=pass dkdmain=klu.ac.in dmarc=pass fromdomain=klu.ac.in); spf=pass (google.com: domain of 2024cse+bnbcccyx7xjleikvjfw2qdbubhjhj4@klu.ac.in designates 209.85.220.69 as permitted sender) smtp mailfrom=2024cse+bnbccCYX7XJLEIKVJFW2QDBUBHLJHSJ4@klu.ac.in; dmarc=pass (p=QUARANTINE sp=NONE dis=NONE) header from=klu.ac.in; dara=neutral header i=@klu.ac.in

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10. Document and Report Findings

- Document all your findings, especially if you detect phishing or spoofing.
- Report suspicious emails to your IT department or email provider for further investigation.

gmail login - Search

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Original Message

mxtoolbox - Search

Email Header Analyzer, RFC822 P...

https://mxtoolbox.com/Public/Tools/EmailHeaders.aspx?huid=79c50387-61a8-4ddf-b393-522e346a43d3

Return-Path	<2024cse+bnbccCYX7XJLEIKVJFW2QDBUBHLJHSJ4@klu.ac.in>
Received-SPF	pass (google.com: domain of 2024cse+bnbcccyx7xjleikvjfw2qdbubhjhj4@klu.ac.in designates 209.85.220.69 as permitted sender) client-ip=209.85.220.69;
Authentication-Results	mx.google.com: dkim=pass header i=@klu.ac.in header s=google header b=QxEfXkI; arc=pass (i=3 spf=pass spfdomain=klu.ac.in dkim=pass dkdmain=klu.ac.in dmarc=pass fromdomain=klu.ac.in); spf=pass (google.com: domain of 2024cse+bnbcccyx7xjleikvjfw2qdbubhjhj4@klu.ac.in designates 209.85.220.69 as permitted sender) smtp mailfrom=2024cse+bnbccCYX7XJLEIKVJFW2QDBUBHLJHSJ4@klu.ac.in; dmarc=pass (p=QUARANTINE sp=NONE dis=NONE) header from=klu.ac.in; dara=neutral header i=@klu.ac.in
DKIM-Signature	v=1; a=rsa-sha256, c=relaxed/relaxed, d=klu.ac.in, s=google, t=1787662892, x=1788267692, darn=klu.ac.in, h=list-unsubscribe:list-archive:list-help:list-post:list-id :mailing-list:precedence:x-original-authentication-results:x-original-sender-content-type:subject:message-id:date:from:in-reply-to:references:mime-version:version:x-gm-gg-x-beenthere:x-gm-message-state:from:cc:subject:date:message-id:reply-to:content-type; bh=xamsobcbg+24HJQFSIO9vaVHnkH49/GRizniCuWAaM=-; b=QxEfXkIq+x49ZLco6ulOkP6sW3WsiG0UBsuJKBqn7wQx3uXx4MSXRUgkieryyQA xWhNp1tqPTPdPYOOekhpRLX1nu0rPUXpZE0282Zry6W5sQCYEQ7t+FEsLlnCA4N1Lvc :i85D4fT1aKbtlq22nnG3CBxp4N2vk1Cz9sV7p45F7kPShL01YUSRYXr6efpVDM rjXFey13PgpDG3bLAnpmlK2zcPyAzAxSx2JJeFhySSVYByzZGowQ3gadT4EPy6QMb JX8uIB/S75CqQTMbVWUUn3hEdE5b8gdaSxg/biKM6+gSEUv1Dy9f4nKHMMWqj8S7iLLew==
X-Google-DKIM-Signature	v=1; a=rsa-sha256, c=relaxed/relaxed, d=1e100.net, s=20251104, t=1787662892, x=1788267692, h=list-unsubscribe:list-archive:list-help:list-post:x-spam-checked:in-group:list-id:mailing-list:precedence:x-original-authentication-results:x-original-sender-content-type:subject:message-id:date:from:in-reply-to:references:mime-version:x-gm-gg-x-beenthere:x-gm-message-state:from:cc:subject:date:message-id:reply-to:content-type; bh=xamsobcbg+24HJQFSIO9vaVHnkH49/GRizniCuWAaM=-; b=Tie/HHDUrslUbA0k6Ym0k3gQ4VqXWJ0sEgqlgmWkxgJAKARU/AJktyO6CCDFyFA3E mNP1ULEr920XJ9e7sBgptlw0VGygTqBUG+VhCzzIA00WtYTsGsJO22HzTerXbjMVD9Tjm LwktHXcvPmvWPCfaLIF68x20f7wKhZfkqKZoNo5jfbE28cUpGagO/uKQ1t0L3u3a9 BfP6S6mVXxZyGU9xEL2XsH/CWbe99yA541UXLMMRVr28diB0WW1019AX8XKQPCYfckBp UwjokATRpxcdFD81U/omoYq2+of1sU+cJc4Mnx6RdRrA3BkM9P19GZAOWYojhr2SCSk MA6g==
X-Gm-Message-State	AfUf+++mDjxuYq78mS1BrOwSwwogYBkk9A4dUQKUxth+B295EMFUI1ZnR XHIPHQHwEkuELFht37MhNNWyoKnfY1Y662Q89M4QonzpBzoGKb+bR1OgaHcyKvUnleyQ/
X-BeenThere	2024cse@klu.ac.in; h="ATskLdfoHAQsIYnMWecynlUkA+2S2LL5FGRD7cLJA5C8d8eDx0g=="
X-Gm-Gg	AR++d114KslF1RMi+Pym4wcvAmTPBshdMRIDlqkZFclwUHIUIBkCCoJ3Wniz 5f1JKH3o7g1ZxGSobHbmZyhH968zAwWFGS2Lj4OQjQhY042MRZns3eCTp2XieqFV+v2WwbZN4 KSn47nxBAzILY6uhuD80PuaNSV2alu78QOx5f77b17N10d+gHqHxmCS8Y50R5c97NdW1 e+8HqNp8D6k6sPcGnQg3o+PnwfmJWDH46JsmY2bPQSlg5AEq3kb1b4ELMYUNFX7Myt6SCz6jl n4KnmPkhVpc2b9gpp002MozfkrdC0gaMMFDoSKAUe7b6FnySiRMEOy4jDMp570RPV0HB0 yjm3JNSA1Rk0FInmUjC5L1Xqtd4KnnH5RwRyuaEJzUPFoAPCYn5MVJUwmzTpA0Blr7H7T05 SjnhVumwgs3pcnRupNkV6Lv3iq
MIME-Version	1.0
References	<CABnRfQE+8TfaFwnSRssdP00k5n306OJYrm=v5-C4W6C=9uxTzA@mail.gmail.com>
In-Reply-To	<CABnRfQE+8TfaFwnSRssdP00k5n306OJYrm=v5-C4W6C=9uxTzA@mail.gmail.com>
From	"Dr. R. Raja Subramanian HOD-CSE" <hodcse@klu.ac.in>
Date	Tue, 25 Aug 2026 18:31:12 +0530
X-Gm-Features	AcwNNIXVbeUCAnm8RygJAlpKXY2fI3E0IE4Zi89nORi5vF7DR94Dm9ulwgjVs
Message-ID	<CAMq6mVbZKH-MSBe-cAKSLSC1EU41MSv06e5-j5=cObjpLXg@mail.gmail.com>
Subject	Fwd: GATE 2027 & JAM 2027 Examinations - Reg

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Example of a Header Analysis

Here's a quick example with a sample header:

vbnet

Copy code

Received: from mail.example.com (mail.example.com [192.0.2.1])

by mail.receiver.com with ESMTP id u29si8604336pjs.40.2023.08.10.07.00.16;

Thu, 10 Aug 2023 07:00:16 -0700 (PDT)

Received: by mail.example.com with SMTP id a1mr1243772ywh.51;

Thu, 10 Aug 2023 07:00:15 -0700 (PDT)

Message-ID: <CA+7eu=4pSeXgQ@mail.example.com>

- **Check IP Addresses:** Lookup 192.0.2.1 and ensure it is associated with mail.example.com.
- **Verify SPF/DKIM:** Check if mail.example.com is authorized to send emails for this domain.
- **Compare Time Stamps:** Ensure the times make sense chronologically without unusual delays.